

Civics Group Reg Number

Candidate's Name _____

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MERIDIAN JUNIOR COLLEGE

JC2 Preliminary Examinations

Higher 1

H1 Biology

Paper 1 Multiple Choice

8875/01

26 September 2014

1 hour

Additional Materials: OMR

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and register number in the spaces at the top of this page and all separate paper used.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMR answer sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

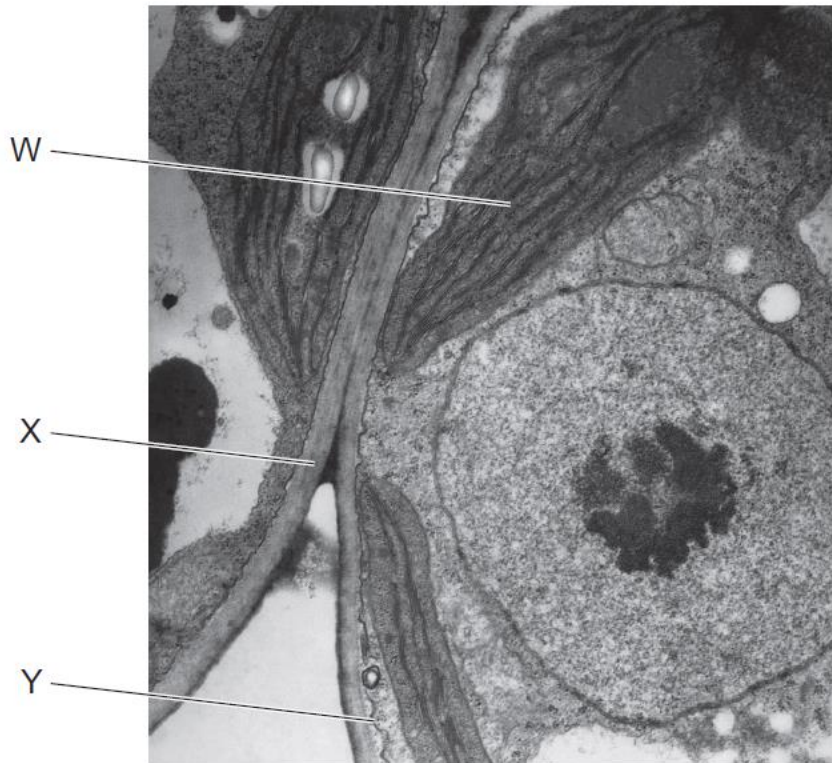
This paper consists of **17** printed pages

[Turn over]

1. Which type of cell will contain the highest proportion of single membrane-bound structures?

- A** Post synaptic neurone
- B** β cell of Islets of Langerhans
- C** Red blood cell
- D** Smooth muscle cell

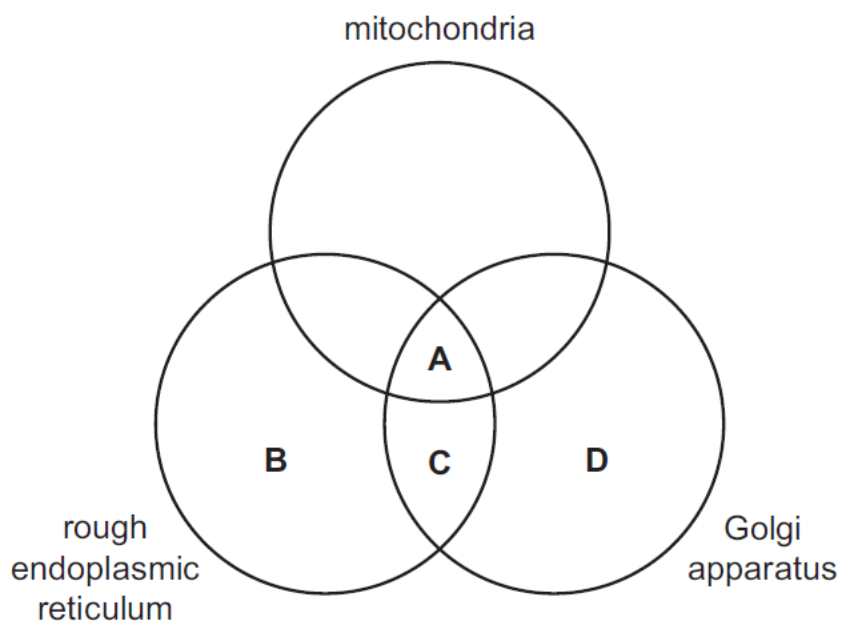
2. The electron micrograph shows part of two cells.



Which of the labelled features enable these two cells to be identified as eukaryotic?

- A** X only
- B** W only
- C** W and X
- D** W, X and Y

3. Which organelles are required in the formation of glucagon found in secretory vesicles?



4. Four different fruit juices, **A**, **B**, **C** and **D**, were tested with Benedict's solution. A second sample of each juice was hydrolysed and tested with Benedict's solution. The table shows the masses of the precipitates formed.

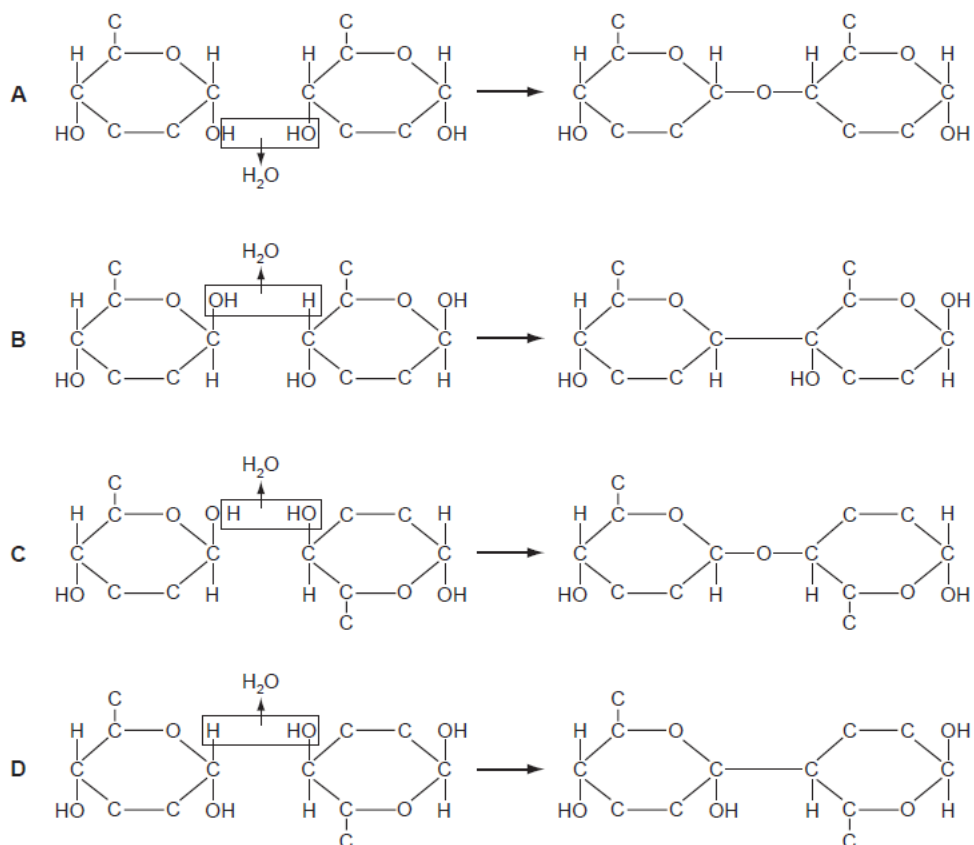
Which juice contains the greatest mass of non-reducing sugar?

	mass of precipitate before hydrolysis / mg	mass of precipitate after hydrolysis / mg
A	60	82
B	52	70
C	63	84
D	56	77

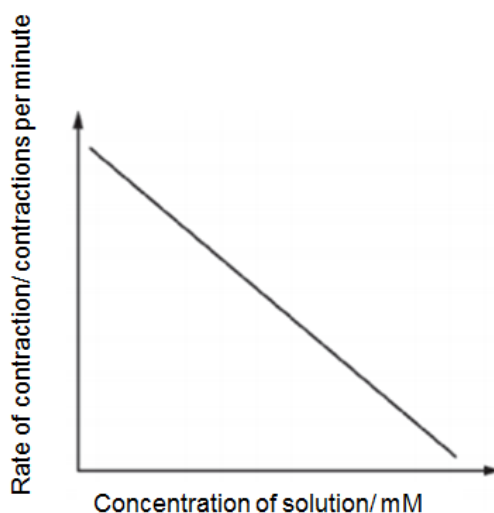
5. Which set of statements correctly describes haemoglobin?

A	four polypeptide chains, forming a globular chain	Iron ions can associate with oxygen forming oxyhaemoglobin	In each chain, hydrophilic R groups of amino acids point towards the centre of the molecule	at 50% saturation, two oxygen molecules are transported by the molecule
B	four polypeptide chains, each containing a prosthetic group	iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	at 50 % saturation, two oxygen molecules are transported by the molecule
C	four polypeptide chains, each containing a prosthetic group	iron ions in the molecule can bind reversibly with oxygen	in each chain, hydrophobic R groups of amino acids surround the iron ion	at 50% saturation, two oxygen atoms are transported by the molecule
D	polypeptide chains produce a loose helical shape, which curls to form a spherical molecule	Iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	at 50% saturation, two oxygen molecules are transported by the molecule

6. Which diagram shows the linking of two monomers together to form cellulose?



7. Paramecia are unicellular protists that have contractile vacuoles to remove excess intracellular water. In an experiment, paramecia were placed in salt solutions of increasing concentration. The rate at which the contractile vacuole contracted to remove excess water was determined and plotted against concentration of solutions, as shown in the graph. Which of the following is the correct explanation for the data?



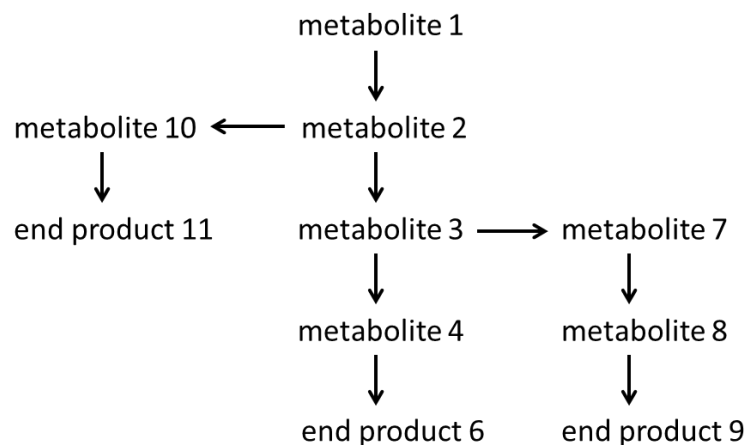
- A** At higher concentration of solution, lower rates of contraction are required because more salt diffuses into the paramecia.
- B** As the concentration of solution decreases, the contraction rate increases because the amount of water entering the paramecia by osmosis increases.
- C** The contractile vacuole is less efficient in solutions of high concentration because of the reduced amount of ATP produced from cellular respiration.
- D** In isotonic salt solution, there is no movement of water into or out of the paramecia, so the contraction rate is near zero.
8. What supports the view that a membrane protein is involved in active transport?
- A** It allows movement of molecules across a membrane if concentration differences exist.
- B** It can only function if mitochondria are supplied with sufficient oxygen.
- C** It has a tertiary structure with a binding site with a specific shape.
- D** It is found in the cell surface membranes and the mitochondrial membranes.

9. Potassium cyanide is known to interfere with the formation and use of ATP in cell metabolism.

If the use of potassium cyanide resulted in an accelerated entry of a solute into a cell, it may be reasonably assumed that, under normal circumstances, the solute enters by

- A active transport
- B passive diffusion
- C osmosis
- D pinocytosis

10. The diagram below shows a metabolic pathway that is controlled by end product inhibition. When the concentration of the end products increase beyond a particular level, an enzyme in the pathway is inhibited to lower the rate of production.

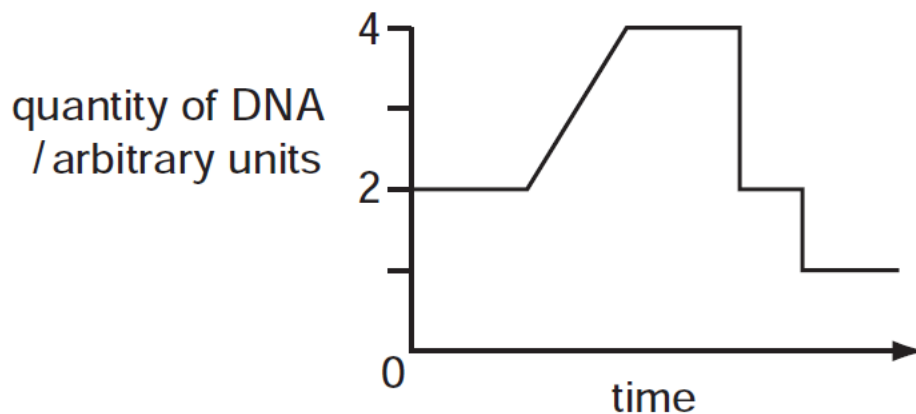


It was discovered that an increase in amount of end product 9 does not affect the synthesis of end product 11, however, it decreases synthesis of end product 6.

The enzyme controlling which of the following reactions could be inhibited by end product 9?

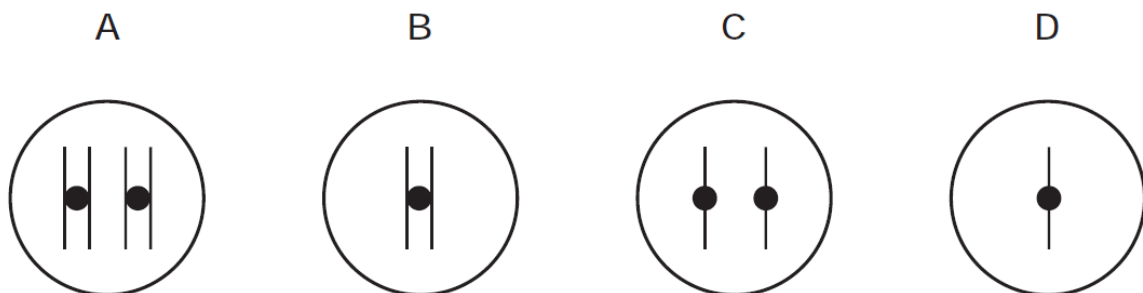
- A metabolite 1 → metabolite 2
- B metabolite 2 → metabolite 3
- C metabolite 3 → metabolite 7
- D metabolite 2 → metabolite 10

11. The graph shows the change in the quantity of DNA in a cell during a reduction division (meiosis).



A cell with one pair of chromosomes ($2n = 2$) undergoes meiosis.

Which nucleus is formed as a result of this division?



12. When DNA replicates, new nucleotides containing the common isotope of nitrogen (^{14}N) are used to build new nucleic acids.

In the laboratory, nucleotides can be synthesised using the heavy isotope of nitrogen (^{15}N). DNA is allowed to replicate once using these ^{15}N nucleotides, then twice more using ^{14}N nucleotides.

What will be the ratio of ^{14}N to ^{15}N nucleotides in the final molecules?

- A** 1:1 **B** 3:1 **C** 5:1 **D** 7:1

13. The table shows some aspects of DNA replication in a prokaryotic cell.

Which combination is correct?

	Leading strand synthesis	Lagging strand synthesis	Direction in which template is read	Primers
A	Towards replication fork	Away from replication fork	5' → 3'	DNA primers
B	Towards replication fork	Okazaki fragments present	3' → 5'	RNA primers
C	Okazaki fragments present	Towards replication fork	5' → 3'	DNA primers
D	Away from replication fork	Okazaki fragments present	3' → 5'	RNA primers

14. Two enzymes, X and Y, are each encoded by different alleles of the same gene. The amino acid sequences of the two enzymes differ between positions 87 and 91 of the polypeptides.

The amino acid sequences of enzymes X and Y, and the corresponding DNA sequence of enzyme X from position 86 to position 93 of the polypeptides, are shown in the table below.

	←N terminal end amino acid position C terminal end→							
	86	87	88	89	90	91	92	93
DNA triplet codes for enzyme X	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA
Amino acid sequence of enzyme X	lys	ser	pro	ser	leu	asn	ala	ala
Amino acid sequence of enzyme Y	lys	val	his	his	leu	met	ala	ala

The actual mRNA codons for amino acids in these positions for enzymes X and Y, are shown in the table below.

Amino acid	lys	ser	pro	leu	asn	ala	val	his	met
mRNA codon(s)	AAA	AGU UCA	CCA	CUU UUA	AAU	GCU	GUC	CAU CAC	AUG

What could account for the difference in amino acid sequence of enzymes X and Y?

- A** A single frame shift deletion in the DNA code at position 87.
- B** Frame shift mutations in the DNA codes at position 87 and position 90.
- C** A change in the sequences of the second and third nucleotides at positions 87 and 88 of the DNA codes and frame shifts at position 89 and 91.
- D** A deletion in the DNA code at position 87 and an insertion into the DNA code at position 92.

15. Two genes, Q and R, affect the size of the petals of a flower.

Gene Q has two alleles, Q^L and Q^A . The genotype Q^LQ^L produces large petals, Q^LQ^A produces small petals, and in Q^AQ^A , petals are absent.

Gene R has two alleles. R produces a red pigment and is dominant over the allele r that produces no pigment.

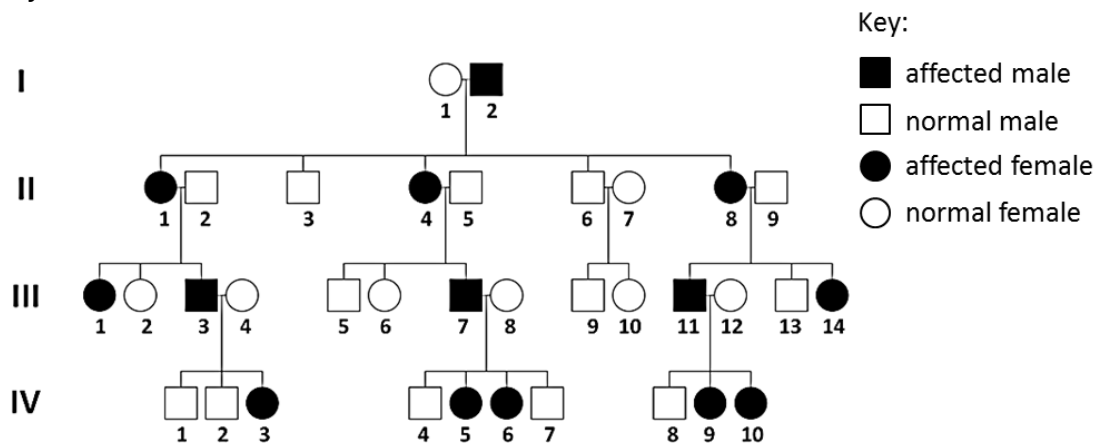
Two plants, both heterozygous for both genes, are crossed.

How many phenotypes are expected in the next generation?

- A** 4 **B** 5 **C** 9 **D** 12

16. Hereditary hypophosphatemic rickets is a genetic disorder that results in low level of phosphate in the blood (hypophosphatemia).

The figure below shows the inheritance of this disease over four generations in an extended family.



Which of the following explains the mode of inheritance of hereditary hypophosphatemic rickets?

- A** Autosomal dominant
B Autosomal recessive
C Sex-linked recessive
D Sex-linked dominant

17. Mendel crossed plants homozygous for yellow, round seeds with plants homozygous for green, wrinkled seeds. All the first generation (F1) had yellow, round seeds. Mendel then crossed two F1 plants.

What proportion of the offspring would be homozygous for one of the recessive alleles?

- A** $\frac{1}{8}$ **B** $\frac{1}{4}$ **C** $\frac{3}{8}$ **D** $\frac{1}{2}$

18. The statements give information about genetic inheritance.

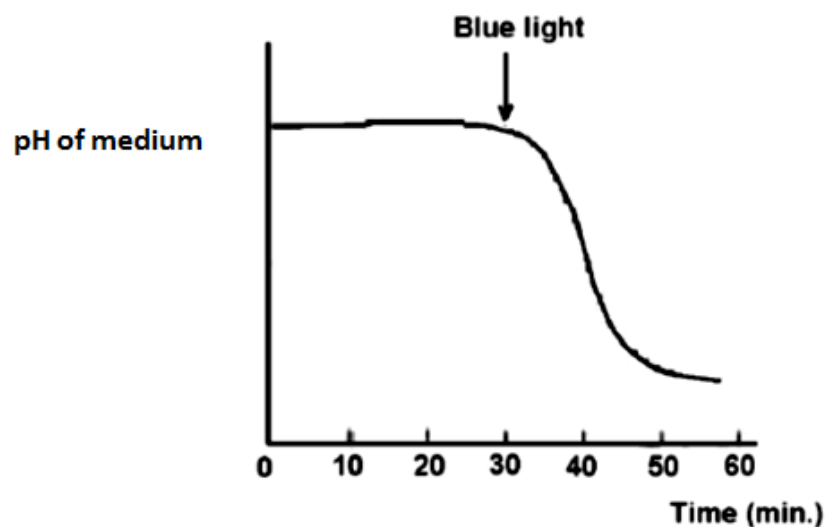
- 1 At fertilization an organism inherits one set of chromosomes from each parent. Each set carries one of each type of gene.
- 2 Each set of chromosomes carries specific genes at specific places.
- 3 The same gene may exist with different DNA sequences due to mutation.
- 4 Individuals may have the same DNA sequences for both copies of a gene.
- 5 Individuals may have two different DNA sequences of a gene.
- 6 The expression of some versions of genes is hidden by the expression of other versions of the same gene.

	1	2	3	4	5	6
A	alleles	loci	genotype	homozygous	heterozygous	dominant
B	loci	genotype	heterozygous	alleles	homozygous	recessive
C	genotype	loci	alleles	homozygous	heterozygous	recessive
D	loci	homozygous	genotype	alleles	dominant	heterozygous

19. What happens to the reduced NAD molecules in cell metabolism?

- A** They act as oxidising agents in glycolysis.
- B** They are oxidised in thylakoid membrane for ATP formation.
- C** They are oxidised in the Calvin cycle when react with glycerate-3-phosphate.
- D** They are oxidised in the mitochondria when react with pyruvate.

20. The result of an experiment which uses guard cell protoplasts of *Vicia faba* is given below. Protoplasts (plant cells without cell wall) were incubated in a suspension medium with isotonic osmotic pressure. After 30 min under saturating red light they were irradiated with blue light for 30 sec. During the experiment in which the protoplasts were cultured the pH of the medium was monitored.



What would be the most plausible conclusion based on the above results?

- A** Blue light may help guard cells to take up protons from outside into the cell.
- B** Blue light may enhance the ability of guard cells to pump protons out of the cell.
- C** Blue light may be a very effective wavelength of light for the respiration of the guard cells.
- D** Other wavelengths of light may help guard cells to transfer protons.

21. What is required for ATP synthesis in mitochondria?

- A** Active pumping of protons into the matrix
- B** Diffusion of protons out of the matrix
- C** Accumulation of protons in the intermembrane space
- D** Accumulation of protons in the matrix

22. Plant chloroplasts are believed to have evolved from cyanobacteria-like organisms through endosymbiosis. Genes in the cyanobacterial ancestor are retained in chloroplast genomes. Which of the following statements support this hypothesis?

- I. Chloroplasts and cyanobacteria share similar photosynthetic pigments and thylakoid membranes.
- II. Cyanobacteria exhibit photosynthesis.
- III. Chloroplasts and cyanobacteria share similar DNA and 80S ribosomes.
- IV. Prokaryotic genes can be expressed in chloroplasts.

- A** I and II only
- B** I, II and IV
- C** I and III only
- D** I, II, III and IV

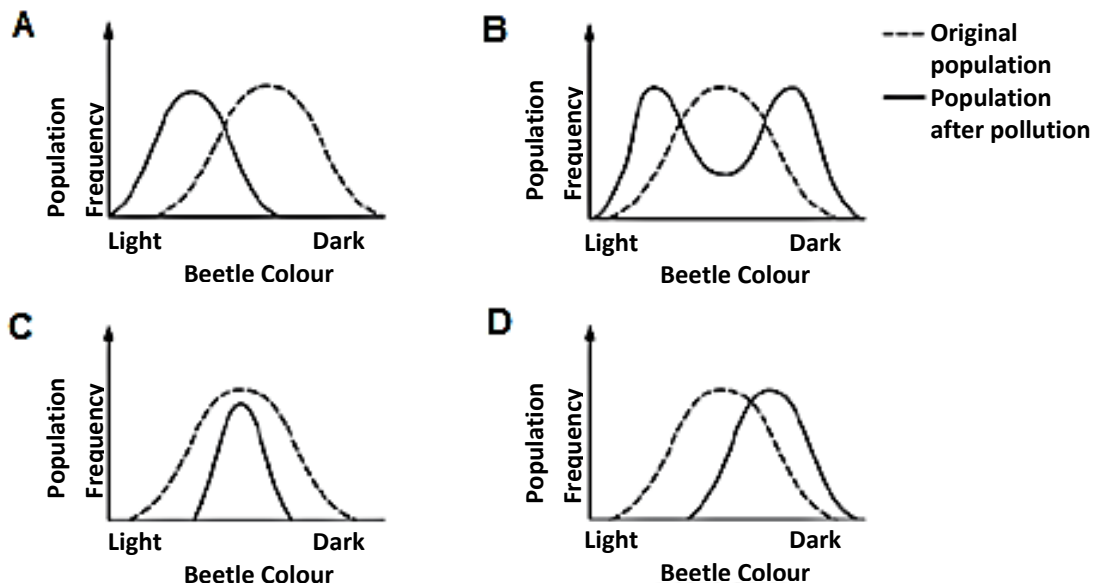
23. Which of the following statements are true for natural selection?

- I. The mechanism is random.
- II. It usually affects the smaller population.
- III. It results in adaptation.
- IV. It affects the genetic make-up of the population.

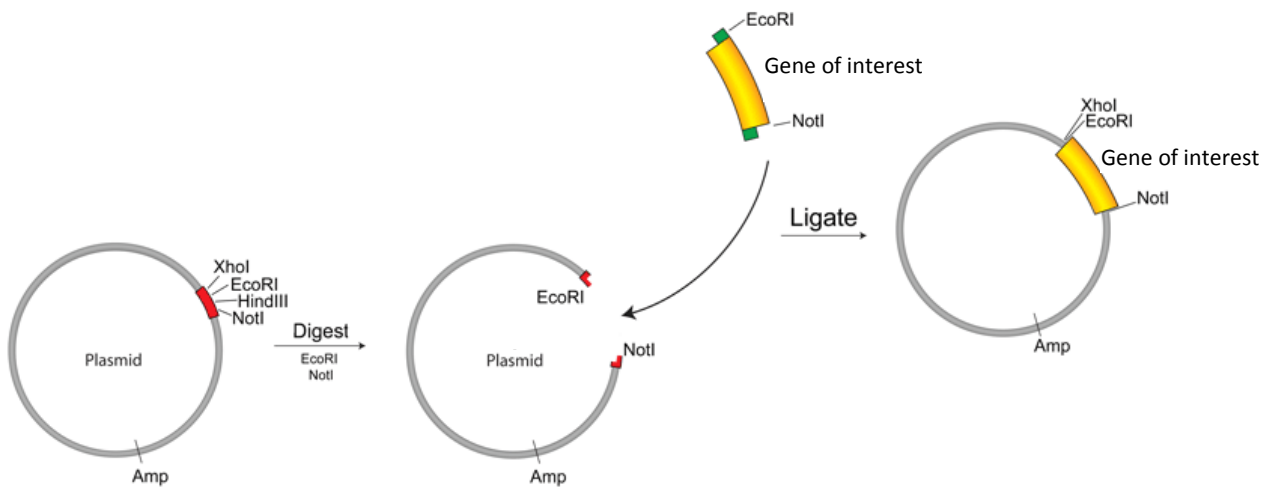
- A** I and IV
- B** I, III and IV
- C** II and III
- D** III and IV

24. In a hypothetical population of beetles, there is a wide variety of colour, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.

Which of the following graphs includes the most likely change in the coloration of the beetle population after pollution?



25. Diagram below shows a process of genetic engineering.



All of the following statements incorrectly explain the purpose of the process, except

- A** The gene of interest was inserted into the genetic marker to allow for identification of bacteria with recombinant plasmid.
- B** The recipient plasmid was isolated from a bacteria and grown in mass to ensure success of this process
- C** The use of two separate restriction enzymes to cut the plasmid and gene of interest allows the gene of interest to be inserted in a specific orientation.
- D** *EcoRI* and *NotI* restriction sites are chosen to prevent insertional inactivation of *Amp* gene.

26. Which of the 2 primers can be used to amplify the sequence shown below by Polymerase Chain Reaction?

5' – GGAATTCGT -- // -- TATGCTACC – 3'

- A** 5' GGTAGCATA 3' & 5' GGAATTCGT 3'
- B** 5' ACGAATTCC 3' & 5' TATGCTACC 3'
- C** 5' TATGCTACC 3' & 5' GGAATTCGT 3'
- D** 5' ACGAATTCC 3' & 5' GGTAGCATA 3'

27. Identification of individuals predisposed to a disease is done by performing DNA analysis techniques such as Polymerase Chain Reaction and gene probing after Southern Blotting. PCR requires the presence of primers for amplification of target gene while a gene probe is used to identify gene of interest.

Which of following comparison of gene probe and primers are false?

- I. Both are short and single stranded nucleic acids.
- II. Both have sequences that are complementary to 3' end of the target gene.
- III. Both can only be made up of deoxyribonucleotides.
- IV. Both must be radioactively-labelled to be identified by autoradiography.

- A** I and II
- B** II and IV
- C** III and IV
- D** II, III and IV

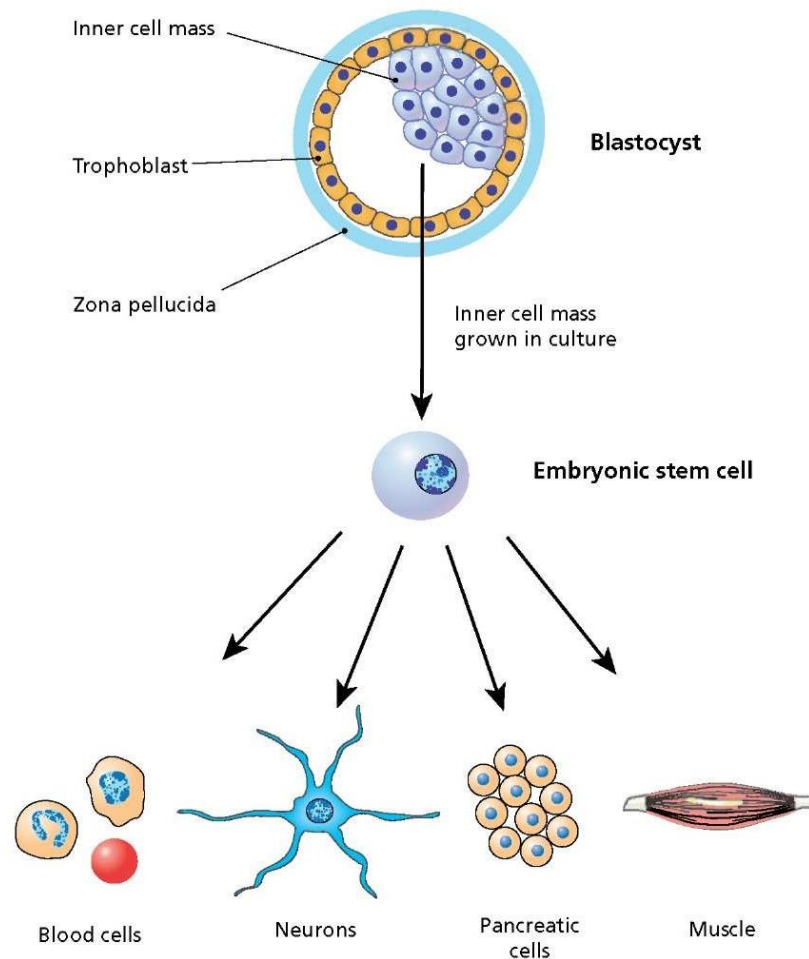
28. Some stages of a procedure involved in paternity testing are listed.

- 1 Alkali denatures DNA
- 2 Fragments are migrates at different rate depending on the number of base pairs
- 3 Probes bind to DNA which can be identified.
- 4 DNA samples are obtained from different individuals
- 5 Fragments are transferred from agarose gel to nylon membrane.
- 6 Several STR loci are isolated using suitable restriction enzymes

Which sequence shows the correct order of these stages?

- A** 6 → 4 → 2 → 5 → 1 → 3
- B** 6 → 4 → 2 → 1 → 5 → 3
- C** 4 → 6 → 2 → 5 → 1 → 3
- D** 4 → 6 → 2 → 1 → 3 → 5

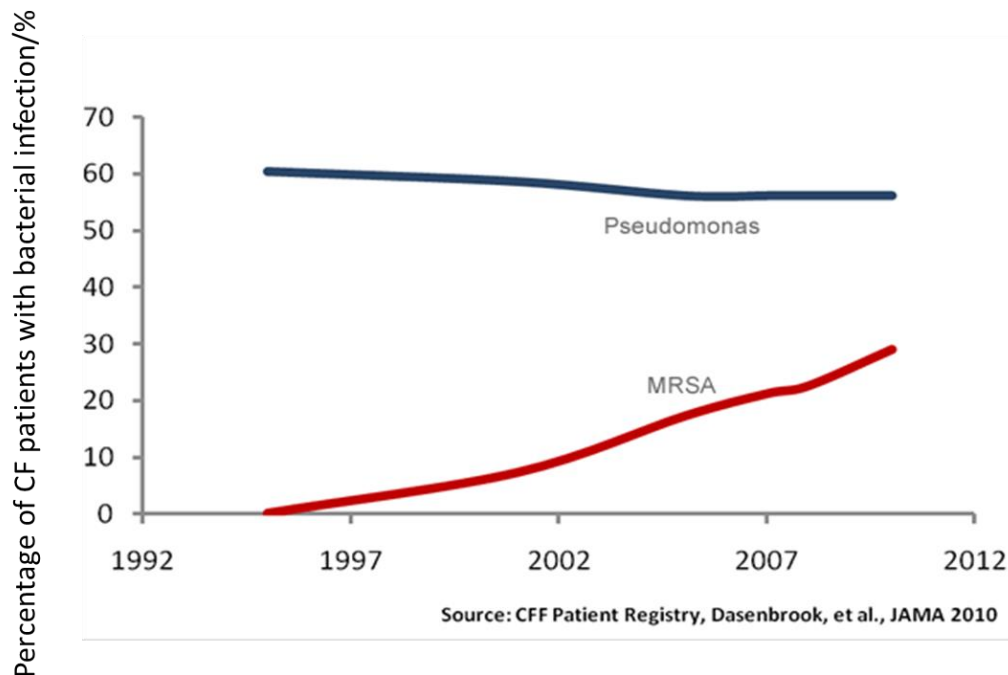
29. Which of the following features of the embryonic stem cells and specialized cells shown in the diagram are true?



	Embryonic stem cells	Specialized cells
A	Embryonic stem cells display greater plasticity when grown in culture than when in blastocyst.	The blood cells are genetically different from the embryonic stem cells but have shorter telomeres.
B	Embryonic stem cells are pluripotent and are capable of transdifferentiating into many different cell types.	The pancreatic cells are genetically identical to the embryonic stem cells but with a different set of genes expressed.
C	Embryonic stem cells are multipotent and are capable of differentiating into limited range of cell types.	The blood cells are genetically different from the embryonic stem cells because different genes are expressed.
D	Embryonic stem cells are pluripotent and are capable of differentiating into many different cells types.	The pancreatic cells are genetically identical to the embryonic stem cells but have shorter telomeres.

30. Cystic fibrosis (CF) is a genetic disease characterized by thick, sticky mucus in the lungs and chronic lung infections resulting in gradual loss of lung function. The most prevalent lung pathogen in CF patients is *Pseudomonas aeruginosa*, which is commonly treated using inhaled antibiotics.

In recent years, methicillin-resistant *Staphylococcus aureus* (MRSA), a bacterium that is resistant to conventional antibiotics has become increasingly common. Persistent MRSA infections have been found to cause a decline in lung function and are associated with a significantly shortened life expectancy. Currently there is no approved inhaled treatment for MRSA infections in CF patients.



Which of the following conclusions can be derived from the information provided?

- I. The inhaled antibiotics against *Pseudomonas* are ineffective which results in the high percentage of CF patients infected with *Pseudomonas*.
 - II. The inhaled antibiotics against MRSA are ineffective which result in an increase in percentage of CF patients infected with MRSA.
 - III. The thick mucus in CF obstructs the pancreatic duct and prevents pancreatic enzymes from reaching the small intestines to digest the bacteria.
 - IV. *Staphylococcus aureus* is likely to contain gene that codes for intracellular proteins that reduce antibiotic inhibitory effect.
- A II only
- B IV only
- C I, II and IV only
- D II, III and IV only

END-OF-PAPER